

A FAMILY PROPERTY RECORDS SYSTEM

CASE STUDY: KAMPALA, UGANDA

BY

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**A RESEARCH REPORT SUBMITTED TO THE
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DECLARATION

I, **Kobusingye Muramuzi Lunah**, declare that the work presented in this research project report is my original work and has not been submitted to any University or Institution of Higher Learning for any academic award. All work from other authors has been fully and properly acknowledged and cited.

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DEDICATION

This research is dedicated to the families of Uganda, whose commitment to preserving their history and securing their future through record keeping serves as a testament to their strength and resilience. May this work contribute to the protection of family legacies, ensuring that every voice is heard and every right is upheld for generations to come.

To my father, Mr. Muramuzi Frank and my brothers, Emma and Onesmus, whose unwavering support has been my greatest source of strength. To my late mother, Mrs. Muramuzi Patience, whose love and wisdom continue to guide me in spirit—this work is a tribute to all of you. You have been my pillars, and I am forever grateful.

ACKNOWLEDGMENT

I extend my deepest gratitude to my family, whose unwavering support has been my anchor throughout this journey. To my father, Mr. Muramuzi Frank, thank you for your endless support. To my late mother, Mrs. Muramuzi Patience, your love and wisdom continue to inspire me every day—this achievement is as much yours as it is mine.

To my mentors, friends and colleagues, your words of encouragement and shared moments of motivation were vital in seeing this project through. A special thanks to the community members of Kampala, Uganda and participants who generously shared their stories; your voices are the heart of this work.

This project stands on the shoulders of many, and I am humbled by the collective effort that brought it to life.

LIST OF ACRONYMS

This is a list of acronyms used in this report.

LIS: Land Information System
NLIS: National Land Information System
UGX: Ugandan Shilling
DBMS: Database Management System
IT: Information Technology
ICT: Information and Communication Technology
RDBMS: Relational Database Management System
UBOS: Uganda Bureau of Statistics
PHP: Hypertext Preprocessor
HTML: Hypertext Markup Language
CSS: Cascading Style Sheets
MySQL: My Structured Query Language
KII: Key Informant Interview
IDI: In-Depth Interview
ID: Identification
fps: family property system

DEFINITION OF TERMS USED

These are the definitions of terms that are used in this report;

- **Family:** A family is a group of individuals related by blood, marriage, or legal ties who share a common household and responsibilities toward one another. In the context of property management, it refers to those who have legal rights or interests in family-owned property.
- **Family Property Record System:** This is a centralized digital platform designed to store, manage, and access family property-related documents and information. This system aims to improve record accuracy, reduce disputes, and enhance overall property management efficiency.
- **User Profile:** This is a structured record that contains essential information about an individual accessing the system, including their name, contact details, role (e.g., family member or administrator), and access permissions. It helps manage user access to sensitive property information.
- **Property Record:** This is an entry that contains detailed information about a specific property, including its unique Property ID, location, type, ownership history, and any associated legal documents such as title deeds. It serves as the central data point for managing family assets.
- **User Authentication:** This is the process of verifying a user's identity before granting access to the system, typically through unique credentials like a username and password. This process is crucial for protecting sensitive information and ensuring that only authorized users can access the system.
- **Data Migration:** This is the process of transferring existing property records from the old manual system to the new digital platform. This process includes cleaning and validating data to ensure accuracy and consistency in the new system.
- **Property Fraud:** This refers to illegal activities involving the misrepresentation of property ownership or rights, often leading to financial losses for legitimate owners. The Family Property Record System aims to reduce the risk of property fraud through improved documentation and verification processes.
- **XAMPP:** XAMPP is a free and open-source cross-platform web server solution stack package that simplifies the installation of Apache, MySQL, PHP, and Perl. It is commonly used in the development of web applications, including the Family Property Record System, allowing developers to create and test their applications locally before deploying them.

- **JavaScript:** JavaScript is a programming language used primarily for creating interactive and dynamic content on websites. In the context of the Family Property Record System, JavaScript is employed to enhance user experience by enabling features such as real-time form validation, dynamic content updates, and interactive user interfaces.

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ABSTRACT

This research study investigated the problem of family property recording and management in Uganda, focusing on issues such as fragmented record-keeping and the lack of formal documentation. One of the major problems identified was that families often struggled with property disputes due to outdated and scattered records, which led to legal and financial insecurities. The main purpose of the study was to develop a centralized digital Family Property Record System to improve the management and accessibility of family property documents.

Methods of Systems Analysis and Design (SAD) were used to collect and elicit various requirements, along with gathering relevant data through in-depth interviews (IDIs), and surveys.

One of the major findings of this study was that the developed Family Property Record System effectively centralizes and organizes family property records, improving accuracy and reducing disputes. The system allows users to access important information in real-time and facilitates automated data entry, which enhances record-keeping efficiency. Additionally, the system provides secure access to authorized family members, ensuring that sensitive information is protected.

This study recommends that families and community organizations adopt the developed system to improve property management and record accuracy. Training programs should be established to enhance users' digital literacy, and ongoing support should be provided to ensure the system meets the evolving needs of families in Uganda.

CHAPTER ONE: INTRODUCTION

1.0 INTRODUCTION

This chapter gives an insight on A Family Property Records System. This includes the background of the study, problem statement, aims and objectives of the study, research gaps, research questions, scope of study and the significance of the study.

1.1 BACKGROUND OF THE STUDY

Traditional methods of managing land records involve physical documents such as title deeds and rental agreements, which are susceptible to loss, damage, and unauthorized access. Managing family property records manually can be a challenging task especially for families with dispersed properties and the bereaved families that have lost a family leader in charge of these properties. As families grow and properties change hands through inheritance or sale, the task of keeping accurate property records has become increasingly challenging. Traditionally, these records have been maintained through physical documents like writing on paper and spreadsheets which are prone to loss, damage and unauthorized access easily (Kiggundu, 2018).

Family property record keeping is also a global issue as it is an issue in East Africa and Uganda. Globally, property record systems are inadequate and unpopular in families which contributes to land disputes, hinder economic development, and perpetuate inequalities, especially affecting vulnerable groups like women and marginalized communities (Deininger, 2003).

In East Africa, including Uganda, there is a problem of fragmented family property records and also the access to secure property records is limited hence impacting land tenure security and fueling family disputes over inheritance and property rights (Kiggundu, 2018).

The government of Uganda is transitioning towards digital solutions to address inefficiencies and enhance security in property record keeping hence the development of the Land Information System (LIS) and the National Land Information System (NLIS) to digitize land records and improve accessibility of these records. This has not been fully exercised in Uganda due to scattered and out dated paper property records which makes the migration of this data into the digital systems complex and insecure. The Ugandan community has also not fully embraced digital recording poor attitude to technology.

Therefore, with this proposed digital Family Property Records System, there is an opportunity to streamline the management of property records. The system will securely organize, store, update and manage all documentation related to family-owned properties. This digital system will enable family members and authorized personnel to easily access and update records, ensuring the accuracy and availability of important documents.

This is will promote efficiency, security, accuracy and integrity in family property management hence providing a reliable technology tool for current and future family generations.

1.2 PROBLEM STATEMENT

Approximately 80% of households in Uganda own land with urban land ownership standing at 21.1% compared to rural land ownership at 78.9% (Uganda Bureau of Statistics, 2023). This presents a possibility of social problem of family disputes over inheritance and property rights as well as inefficiencies in property management caused by the fragmented family property records as families keep property records in different places and formats. This results into family feuds, loss of property and in the worst case, family fueled murders.

Furthermore, with only 20% of landowners having formal title (Uganda Bureau of Statistics, 2023) a problem of legal and financial insecurity is presented which is caused by lack of records of formal documentation like title deeds, wills. This increases the risk of property fraud in the family.

The government of Uganda is transitioning towards digital solutions to address inefficiencies and enhance security in property record keeping hence the development of the Land Information System (LIS) and the National Land Information System (NLIS) to digitize land records and improve accessibility of these records.

This has not been fully exercised in Uganda due to scattered and out dated paper property records which makes the migration of this data into the digital systems complex and insecure. The Ugandan community has also not fully embraced digital recording poor attitude to technology.

The proposed solution is to develop a centralized, digital family property records system that will enable family members to store important family property documents online, authorize access to these documents to the right family members to modify the data and improve record accuracy through automated data entry.

The digital system will be centralized, secure, efficient and user friendly hence reducing the problems caused by property fragmentation and lack of records of family formal documents.

1.3 RESEARCH GAP

The gaps in the research on family property record systems are represented in the table below.

Table 1: Research gap

RESEARCH GAP	DESCRIPTION	LITERATURE
Inadequate Documentation systems.	Current property record systems are fragmented and poorly documented, leading to disputes and inefficiency in record keeping. This is due to missing land titles.	(Uganda Bureau of Statistics (UBOS), 2023)
Technological gap.	Many systems rely on out dated, paper record methods which are inaccurate and not transparent.	(Uganda Bureau of Statistics (UBOS), 2023)

1.4 OBJECTIVES

1.4.1 GENERAL OBJECTIVE

To develop a centralized, secure, efficient and user-friendly digital family records system that stores family property related documents and reduces the risk of property fraud of these properties.

1.4.2 SPECIFIC OBJECTIVES OF THE STUDY.

- I. To evaluate the current property record system used by families in Uganda.
- II. To explore cultural influences on family property management in Uganda.
- III. To develop a centralized family property records system in Uganda.
- IV. To test and validate the functionality of the digital family records system in Uganda.

1.5 RESEARCH QUESTIONS

1.5.1 GENERAL RESEARCH QUESTION.

How can a digital family property records system be developed and designed to effectively manage, store and secure family property and the related documents?

1.5.2 SPECIFIC RESEARCH QUESTION.

- I. What methods are currently used by families to record and manage their property and related documents?
- II. How do family structures and cultural beliefs affect the documentation and transfer of family property?
- III. What features are essential for a centralized family property records system?
- IV. How will the performance of the digital family property records system be tested and validated?

1.6 SCOPE OF THE STUDY

1.6.1 SUBJECT SCOPE

The subject scope of this study on developing a family property records system focuses on structuring and securing databases to store diverse family property records; user experience and interface design, ensuring the system is user-friendly and accessible; and security and privacy, involving data encryption and access control to protect sensitive information. This will solve the problem of lack of proper records of family formal documents.

1.6.2 GEOGRAPHICAL SCOPE

This study will take place in Kampala district in Uganda with a sample of less than five families with their properties. This is because 80% of householders own land with only 20% of these landowners, most probably those in urban areas, have title deeds for the land they own (Uganda Bureau of Statistics, 2023).

These document records can be input into the proposed family property record system.

1.6.3 TIME SCOPE

This study is proposed to take a period of 4 months and that is from July 2024 to October 2024.

Table 2: Research study timescope

Month	Activity
July 2024	Submit research proposal.
August 2024	Data Collection and Analysis.
September 2024	Design the online system.
October 2024	Write and submit the research project report.

1.7 SIGNIFICANCE OF THE STUDY

The study of a family property records system will hold a lot of importance in families and Uganda at large for these reasons.

In terms of legal and estate management processes, a digital family property records system will streamline the management of property and property related documents. This approach will reduce errors and inefficiencies commonly associated with manual documentation, leading to more reliable and organized records. Having a digital system will also ensure that the records meet legal standards, making them more reliable and transparent (Kiggundu, 2018). This will kick-start the government process of policy making and governance regarding family property management in the country.

Economically, the system will help in preserving the value of family assets and organizing them to be used as collateral for loans by ensuring accurate and easily accessible records. This is crucial for maintaining clear property ownership and transfer processes. Additionally, by tracking property values and market trends, families can make informed decisions about property investments and sales, contributing to better financial planning and economic stability (Feder, 1999). This will make the family practicing digital property record keeping more financially united and stronger than the family that still records property using paper.

Socially and culturally, a well-organized property records system can enhance family unity by reducing disputes among family members. Clear, accessible, and indisputable records of ownership and inheritance foster trust and harmony within families. Furthermore, documenting family property and documents serves to preserve family history and heritage, which is particularly important in cultures that value ancestral lineage and property (Deininger, 2003). This will make the family practicing digital property record keeping more socially involved and stronger than the family that still records property using paper.

In terms of security and privacy, implementing security measures such as encryption and access controls will protect sensitive family data from unauthorized access and cyber threats. This enhances trust in digital solutions by ensuring that only authorized individuals can access sensitive documents, thus upholding privacy standards and fostering user confidence

in digital record-keeping. This will make the family practicing digital property record keeping more secure and confidential than the family that still records property using paper.

The study emphasizes sustainability by highlighting the environmental benefits of digitizing property records. This reduces the need for physical storage and paper use, contributing to environmental sustainability efforts by minimizing waste. A sustainable records system ensures that property information remains accurate and accessible for future generations, aiding in long-term family planning and asset management. This creates a safe and clean environment in the country.

1.8 CHAPTER SUMMARY

This chapter of this project proposal presents an overview of the necessity and importance of developing a family property records system. It highlights the necessity of such a system, the benefits it promises, the problems the system will be solving and the system's importance to families in Uganda. Therefore, this chapter lays a solid foundation for the proposal, establishing the background of the study, problem statement, objectives of the study, research questions, scope of the study, time scope and significance of the study.

CHAPTER TWO: LITERATURE REVIEW

2.0 INTRODUCTION

This chapter analyses the literature available on property record systems and synthesizes it with the proposal topic of A Family Property Record System and the different themes that arise from the topic as well as the studies available like Property as a family resource, current vs proposed property record system, property inheritance, the role of technology in property recording, the existing property record systems and conceptual framework.

2.1 PROPERTY AS A FAMILY RESOURCE.

2.1.1 DEFINITION OF TERMS.

- **Family:** This is a group of individuals related by blood, marriage or adoption, often living together, sharing resources and responsibilities.
- **Property:** This refers to assets or resources owned by individuals and have legal rights attached to them.
- **Family property:** This refers to assets collectively owned by and family and passed down through generations.
- **Family Property Record System:** A system designed to organise, manage and store family owned property.

2.1.2 PROPERTY AS A FAMILY RESOURCE.

Property is seen as a very significant family asset that supports family goals and stability. Property ownership is not only a legal issue but also a social one, where property serves as a means of economic support, social status, and family cohesion (Mary Ann Glendon, 1981).

Property not only serves as a material asset but also shapes family identities and intergenerational relationships (Deininger, 2003).

This perspective highlights the importance of a property records system that not only tracks legal ownership but also understands the social implications of property within families hence suggesting that proper property records contribute to peace in a family.

2.2 EXISTING VS PROPOSED PROPERTY RECORD SYSTEM

2.2.1 EXISTING PROPERTY RECORD SYSTEM

The existing property record system is mainly manual with land transactions in Uganda still recorded by writing them on paper, which can be time-consuming and prone to errors. Physical records are vulnerable to loss, damage, or manipulation, affecting the reliability of land tenure information. Therefore, land administration system in Uganda suffers from fragmented and inconsistent land records. This fragmentation contributes to ambiguities and disputes over property ownership, particularly in post-conflict and rapidly urbanizing areas (Kiggundu, 2018).

The government of Uganda is advancing towards a centralized digital land registry to consolidate land records into a unified database. This initiative aims to improve data integrity, accessibility, and transparency in land administration (Kiggundu, 2018).

The government of Uganda is transitioning towards digital solutions to address inefficiencies and enhance security in property record keeping hence the development of the Land Information System (LIS) and the National Land Information System (NLIS) which have digitized land records and improved accessibility of these records to a certain extent.

This system has challenges which include; scattered and out dated paper property records which makes the migration of this data into the digital systems complex and insecure, mostly concentrating on land records and leaving out other types of family records like agricultural, residential records. The Ugandan community has also not fully embraced digital recording due to poor attitude to technology of Ugandans.

2.2.2 PROPOSED PROPERTY RECORD SYSTEM

The proposed system, A Family Property Records System, will be designed and enabled to solve the above issues and challenges of the existing property records system in Uganda.

The proposed system, A Family Property Records System advocates for a centralized digital property registry that integrates all family records into a unified platform. This initiative aims to enhance data integrity, accessibility, and transparency in family property management in Uganda. This will enable family members to store important family property documents online, authorize access to these documents to the right family members to modify the data and improve record accuracy through automated data entry.

The system will also store various family property records other than land documents which will include the agricultural and residential records of families for easy management.

Using this system, information is stored electronically, making it quicker to find and update records, reducing mistakes and making it easier for family members to access information from anywhere.

2.3 PROPERTY INHERITANCE.

Property inheritance within families is a complex process that can sometimes lead to conflicts and inequalities within the family. Having proper property records makes the process happen with less or no conflicts (Mary Ann Glendon, 1981).

Accurate and reliable land records are essential for resolving disputes and re-establishing ownership. Therefore, without proper documentation, disputes over land ownership can persist, hindering economic recovery and social stability (Manirakiza, 2014).

With only 20% of landowners having formal land titles in Uganda (Uganda Bureau of Statistics, 2023), a problem of legal and financial insecurity is presented which is caused by lack of records of formal documentation like title deeds, wills. This increases the risk of property fraud in the family.

Understanding these dynamics is important in creating a family records property system that accommodates diverse family arrangements and simplifies property inheritance and succession by properly keeping family records and authorizing access to the records to the right family members.

2.4 THE ROLE OF TECHNOLOGY IN FAMILY PROPERTY RECORDING.

Given a study of Rwanda, advancements such as Geographic Information Systems (GIS) and digital databases have transformed land administration in the country. These tools streamline property records, improving their accuracy and accessibility. This facilitates efficient property management but also supports conflict resolution by providing clear, verifiable property ownership information (Manirakiza, 2014).

Technological innovations such as digital land registries, have the potential to safeguard property rights across generations. Integrating technology in property recording not only enhances administrative efficiency but also reinforces family cohesion by preserving transparent and accessible records.

Presence of barriers such as technological infrastructure gaps, data security concerns, and resistance to digital transitions have slowed down the digitizing process of record keeping in developing communities (Kiggundu, 2018).

Technology plays a crucial role in modernizing family property record keeping, enhancing efficiency and transparency of family records.

2.5 EXISTING PROPERTY RECORD SYSTEMS.

Property record systems are critical for land administration, influencing ownership security, dispute resolution, and economic development. These existing systems vary globally, encompassing traditional manual records to advanced digital solutions.

Uganda's property record system is managed by the Ministry of Lands, Housing and Urban Development (MLHUD). Traditionally reliant on manual records, Uganda is transitioning towards digital solutions to address inefficiencies and enhance security.

The Land Information System (LIS) is a significant modernization effort, supported by the World Bank and implemented by the Ministry of Lands, Housing and Urban Development (MLHUD). It aims to digitize land records and integrate them into a centralized system. This has improved land administration in Uganda by enhancing data accuracy, reducing processing times, and increasing accessibility of land documents to people.

Key land offices in Uganda like the Ministry of Lands, Housing and Urban Development (MLHUD) are being equipped with digital tools to improve efficiency in land registration and information retrieval. The benefits of digital land offices, include faster service delivery and better record-keeping, though challenges in infrastructure and training persist.

2.6 CONCEPTUAL FRAMEWORK

This conceptual framework highlights the key stakeholders involved with the family property record system and outlines the specific benefits they will get from its implementation.

Variables (stakeholders)

- **Government Agencies:** These agencies oversee property administration in the country and policy generation. This system will provide accurate, up-to-date data suitable for policy re-making and planning.
Their actions are independent and lead to changes in how property records are managed hence making them the independent variable.
Attributes: Policy Formulation and Administrative Efficiency.
- **Families and Property Owners:** These hold ownership and rights to family property. The system will reduce family disputes over property boundaries and ownership since the system will organize and collect family property records.
Families and property owners are the ultimate beneficiaries of the Family Property Record System hence making them the dependent variables.
Attributes: Property Ownership and Record management.
- **Academicians:** These conduct studies and provide insights into family property management. This system and research study will enable the academicians to analyse trends and impacts of digital property recording in Uganda.
Attributes: Trend Analysis and Academic Publications.

Relationships

- **Government Agencies <-> Families and Property Owners relationship:** Government Agencies will use data from the system to formulate policies affecting property ownership and management. Families and Property Owners will benefit from these policies and the enhanced efficiency in property administration.
- **Government Agencies <-> Academicians relationship:** Government Agencies will provide access to property records which academicians will use for research and analysis. The insights gained by academicians can, in turn, influence policy formulation.
- **Families and Property Owners <-> Academicians relationship:** Families and Property Owners' property records in the system will be analysed by academicians to study trends and impacts. The findings from such studies will inform better property management practices, policies and updates.

The following is the conceptual framework diagram showing how different stakeholders will work together.

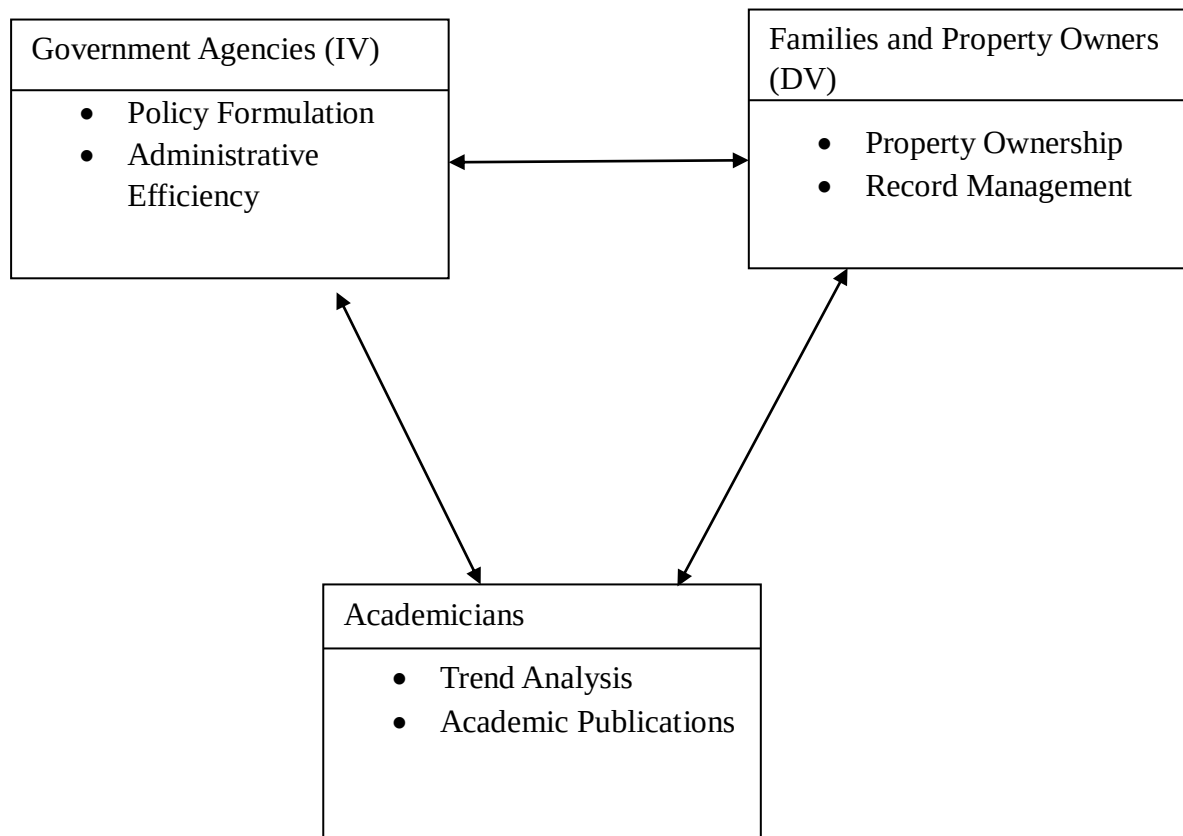


Figure 1: Conceptual framework

KEY:

IV- Independent Variable

DV- Dependent Variable

2.7 CHAPTER SUMMARY

In conclusion, property record keeping has come a long way and it has evolved over time which acts as a strengthening factor for most powerful families in the world. However, it has been realized that the popular family property that has been recorded is land, which has to change as other family properties like houses, cars, businesses and documents like deeds, trusts also need to be recorded as well because they are crucial to the growth of a family.

More-so, future research should focus on addressing the challenges of digitization and exploring ways to integrate family records with broader administrative frameworks to ensure efficiency and security in the system. Educating the public about the benefits and use of the digital land record system is crucial and has to be enforced in Uganda to help in gaining public trust and encouraging widespread adoption of the digital property record systems.

CHAPTER THREE: RESEARCH METHODOLOGY

3.0 INTRODUCTION

This chapter details the research design, population and sample selection, data collection methods, data collection tools, data analysis technique, system requirements and specifications, system design and analysis, system implementation, testing and validation, ethics consideration and chapter summary.

This methodology is guided by the research questions and the specific context in which the Family Property Record system is implemented.

3.1 RESEARCH DESIGN

The following is the research design for this research proposal.

Research Approach

This study employed qualitative, quantitative and mixed methods to provide a comprehensive understanding of the challenges and opportunities in developing a family property record system.

Research Objectives

The general objective of this study is to develop a centralized, secure, efficient and user-friendly digital family records system that stores family property related documents and reduces the risk of property fraud of these properties.

The specific objectives are as follows;

- To evaluate the current property record system used by families in Uganda.
- To explore cultural influences on family property management in Uganda.
- To develop a centralized family property records system in Uganda.
- To test the functionality of the digital family records system in Uganda.

Data Collection methods

This study employs a combination of quantitative with questionnaire method, qualitative with interview method, and mixed-methods like observation method approaches to gather comprehensive insights into property ownership and recording in Uganda. By integrating these methods, the study provides a robust analysis that informs the development of an effective Family Property Record system, tailored to meet the needs of families.

Data Analysis

Collected data through research was analysed by organizing it in Microsoft Word, calculating the population statistics, coding of the interview transcripts to identify recurring themes and patterns as well as transcribing that collected data.

Expected Outcome

The following are the expected outcome of this study;

A detailed understanding of the current property record system in Uganda as well as its shortcomings and acquiring insights into the potential benefits and challenges of implementing the system from multiple stakeholder perspectives.

Timeline

This study took a period of 4 months and that is from July 2024 to October 2024.

3.2 POPULATION AND SAMPLE SELECTION

3.2.1 POPULATION

The population of interest for this study includes individuals and entities involved in family property ownership and management in Kampala, Uganda.

Participants were excluded if they did not meet the defined inclusion criteria or are unable to provide informed consent. Specifically, the study targets:

- **Family heads:** Owners and managers of family property and those maintaining it.
- **Family members and friends:** Individuals that maintain, inherit and expand family property.

3.2.2 SAMPLING STRATEGY

To ensure a representative sample and comprehensive insights, the study will employ the following sampling strategies:

- **Random Sampling:** Within each household, this method was employed by casually selecting participants or family members to represent the whole family to answer questions regarding their experiences with property ownership. This minimized bias and ensure that each subgroup within the population has an equal chance of being represented.
- **Snowball Sampling:** This sampling methods was employed whereby initial participants involved in the research on property records were asked to recommend others who have faced similar challenges in their area, gradually expanding the sample. This is because the network of relationships is crucial for identifying potential participants in this study.

3.2.3 SAMPLE SIZE DETERMINATION

The sample size is determined using the following formula for estimating a population proportion:

$$n = Z^2 \cdot p \cdot (1-p) / E^2$$

Where:

- n = required sample size
- Z = Z-score corresponding to the desired confidence level (e.g., 1.96 for 95% confidence level)
- p = estimated proportion of the population exhibiting the characteristic of interest

- E = margin of error (desired precision)

Calculating Sample Size

- **Estimated Proportion (p):** Based on preliminary data or expert judgment, the proportion of the population likely to exhibit the characteristic of interest will be estimated (e.g., percentage of families involved in property recording).
- **Margin of Error (E):** The acceptable margin of error for estimates derived from the sample will be determined.
- **Confidence Level:** The desired confidence level (typically 95% for social science research) will be selected.

3.3 DATA COLLECTION METHODS

This study employs a combination of quantitative, qualitative, and mixed-methods approaches to gather comprehensive insights into property ownership and recording in Uganda. By integrating these methods, the study will provide a robust analysis that informs the development of an effective Family Property Record system, tailored to meet the needs of families.

3.3.1 QUALITATIVE METHODS

Qualitative method of interviews will be used as an exploratory approach to explore the experiences and perceptions of stakeholders such as family members and property owners on family property recording. In this method for data analysis, the hypotheses are developed after data collection.

The following are the interview type that was exercised;

- **In-depth Interviews:** In-Depth Interviews will purposely explore personal experiences, perceptions, and detailed feedback from a diverse range of family members involved in or affected by property ownership and management issues.

3.3.2 QUANTITATIVE METHODS

Quantitative data on family property records was collected through questionnaires to identify patterns and challenges in property management in the family. It will concern quantifying property ownership patterns, challenges faced by property owners, and the demographic characteristics of bigger families.

Structured questionnaires were used, as an aim to collect specific information that can be statistically analyzed to identify patterns, trends, and relationships related to family. property administration.

3.3.3 MIXED METHODS

Mixed methods procedures employ aspects of both quantitative methods and qualitative procedures. This hybrid approach integrates the best of both methods, and will be done through the observation method.

By directly observing behaviors, processes, and interactions, quantitative data was collected through structured observation using checklists to record specific behaviors and events, such as the frequency and duration of property registration activities related to family-owned properties. Simultaneously, qualitative insights were gathered through unstructured

observation, focusing on interactions between family property owners and officials, and the informal practices affecting family property transactions. This comprehensive approach provided a holistic understanding of the challenges and dynamics within the current family property management system.

3.4 DATA COLLECTION TOOLS

The success of this research proposal hinged on the effective use of diverse data collection tools to gather comprehensive insights into family property records and administration in Uganda. These tools are essential for systematically capturing both quantitative, qualitative and mixed data, ensuring a proper analysis of property management practices and challenges.

By employing a combination of structured questionnaires, observation checklists, and interview guides, the study aims at collecting reliable data from various stakeholders, including property owners, community leaders, and family members. These tools are carefully designed to address the research objectives, enabling a nuanced understanding of the current landscape and informing the development of an efficient Family Property Record system.

3.4.1 QUESTIONNAIRES

To effectively collect quantitative data on family property records and administration in Uganda, this study employed structured questionnaires with predetermined questions regarding family property recording as a primary data collection tool. Each questionnaire included detailed instructions to guide respondents through the completion process, ensuring clarity and consistency.

Online platforms such as Google Forms were used to design and distribute questionnaires electronically via email and social media platforms,

Printed questionnaires were used in areas with limited internet access, ensuring inclusivity and reaching a broader demographic. Clear formatting with easy-to-follow instructions, providing respondents with a straightforward method to provide their input on questions concerning family property records. These were distributed physically using print outs at community centers

The questionnaire has multiple-choice questions on family property recording that will allow respondents to select the most appropriate option(s) from a list, facilitating easy quantification of responses. For example:

“What type of family property do you own?

(a) Residential, (b) Commercial, (c) Agricultural, (d) Other”

It will also include Likert Scale Questions that will measure attitudes or perceptions of respondents on a scale, providing insights into the degree of agreement or satisfaction. For Example:

“Please indicate your level of agreement with the statement:

‘The family property registration process is transparent.’ (1 = Strongly Disagree, 5 = Strongly Agree)”

3.4.2 INTERVIEWS

In order to collect rich qualitative data for the research project on family property records in Uganda, In-Depth Interviews (IDIs) were employed. Typically, the duration per interview was estimated around 30-60 minutes, depending on the depth of the discussion.

The following tools and techniques will facilitate effective data collection through interviews.

- **Interview Guides:** A structured guide with open-ended questions that encourage participants to share their personal experiences, challenges, and expectations related to property registration. The guide will be flexible to allow exploration of emerging themes based on the participant's narrative.
- **Recording Equipment:** Digital voice recorders or recording apps on smartphones/tablets will be used to capture the interview audio, ensuring accurate data collection. Prior to recording, verbal or written consent was obtained from informants, ensuring ethical standards are met.
- **Note-Taking:** Field notes were taken using pen and paper during interviews to capture key points, contextual observations, and non-verbal cues that may not be evident in audio recordings.
- **Arrangements:** Interviews were scheduled at times and locations convenient for informants, whether in person or virtually via video conferencing platforms such as Zoom or Microsoft Teams.

3.4.3 OBSERVATION

This observation method allows researchers to directly witness and record processes, interactions, and behaviors associated with property management, using their eyes and other body senses.

The following tools were employed to ensure systematic and comprehensive data collection through observation method:

- **Observation Checklists:** To guide structured observation by listing specific behaviours, events, or interactions to be observed and recorded. Checklists were tailored to capture relevant quantitative data, such as the frequency of property transactions, the time taken for each registration process, and the number of stakeholders involved.
- **Field Notes:** To capture qualitative data by recording detailed descriptions of observed events, interactions, and contextual factors. Observer took notes on the environment, behaviours, non-verbal cues, and any unexpected occurrences that may influence property management practices, interactions between property owners and officials, informal practices, and any challenges faced during the registration process.
- **Scheduling and Coordination:** Observation sessions were planned and coordinated to cover various locations and contexts, such as family meetings and property recordings. Schedule observations at times when relevant activities are likely to occur, ensuring comprehensive data collection.

3.5 DATA ANALYSIS TECHNIQUE

The data analysis for this research project was conducted to utilize various tools and techniques to ensure thorough and accurate interpretation of the collected data from research. Recommendations based on the analysis, will be provided for improving family property record systems and policies in Uganda.

The following are the techniques that facilitate effective data analysis:

Questionnaire method Data:

- **Data Entry:** Data was input into statistical software like Microsoft Excel and used to clean and organize data collected from structured questionnaires. This involved checking for missing data, outliers, and inconsistencies to ensure data quality and reliability.
- **Data Visualization:** Create charts, graphs, diagrams and tables to visually represent findings and highlight trends or patterns in property administration.

Interview method Data:

- **Transcription of collected data:** Audio recordings, observation and interview notes were converted into text to facilitate thorough analysis and understanding of collected data.

Observation method Data:

- **Data Entry:** Data was input into statistical software like Microsoft Excel and used to clean and organize data collected from observation checklists. This involved checking for missing data, outliers, and inconsistencies to ensure data quality and reliability.
- **Field Notes and Memoing:** Review field notes to identify emerging themes and insights from observed interactions and practices.
- **Thematic Analysis:** Coding of the interview transcripts to identify recurring themes and patterns was done. This will categorize responses into themes related to challenges, expectations, and suggestions depending on the data collected. Analysis of the themes in the context of the broader research questions facilitated the comparison of insights from different types of participants to build a comprehensive understanding of the issues.

3.6 RELIABILITY AND VALIDITY OF RESEARCH INSTRUMENTS

Reliability and validity was reinforced through triangulation, where multiple data sources, such as interviews, surveys, and document reviews, were used to cross-check the information gathered. This helped verify the accuracy of the data and reduced biases. Moreover, expert validation was sought by consulting with professionals in property recording and

management like family leaders, to ensure the research instruments were aligned with industry standards and captured the key issues relevant to family property management in Uganda. This comprehensive approach ensured both the reliability and validity of the data collected for the study.

3.7 ETHICS CONSIDERATION

For this project of a family property records system, ethical approval was very essential. This is because of the need to ensure the protection and respect of participants and the integrity of the research process. Key ethical considerations for this study include informed consent, confidentiality, and minimizing harm.

Informed consent was carefully obtained from all participants physically on paper, ensuring that they are fully aware of the study's purpose, procedures, potential risks, and benefits. This information was provided in languages and formats that are easily comprehensible to the involved parties. For participants under 18, consent will be obtained from parents or guardians, with the child's approval also being sought.

Confidentiality was maintained by anonymizing all data and storing it securely. Interviews will be conducted in private settings to ensure privacy, and participants will be reassured that their responses will remain confidential and they will. Access to data will be restricted to the researcher only, and any publication of results will ensure that individual participants cannot be identified.

A minimizing harm policy was established to protect both the researcher and participants. This will guide the research process to minimize any potential risks or harm to participants. The research will be designed to ensure that participants are not exposed to unnecessary risks, and any foreseeable risks will be communicated to participants before their involvement.

3.8 CHAPTER SUMMARY

This chapter has outlined a systematic and carefully planned research methodology aimed at developing a strong Family Property Record system in Uganda. The methodological framework which includes the research design, population and sample selection, data collection methods, data collection tools, data analysis technique, system requirements and specifications, system design and analysis, system implementation, testing, validation and ethics consideration comprehensively addresses the thorough steps of developing this system.

By adhering to this methodology, the study aims to generate actionable insights and recommendations that can aid practical implementation strategies of the research. Ultimately, the envisioned Family Property Record system seeks to promote transparency, reduce disputes, and empower families by securing their property records effectively.

CHAPTER FOUR: SYSTEM STUDY, ANALYSIS AND REQUIREMENTS ELICITATION.

4.0 INTRODUCTION

This chapter focuses on the systematic examination and analysis of the Family Property Records System detailing the description of the current manual recording system, the weaknesses and strengths of the current system and the requirements of the new system.

4.1 DESCRIPTION OF THE CURRENT SYSTEM.

In the current property record system, when land needs to be registered, property owners are required to visit local land offices to begin the process. Property owners provide physical documents, such as land titles, which are manually recorded by the local land office staff. These records are maintained at various district offices, depending on the location of the property.

The documents submitted are reviewed by land officers for completeness and accuracy. If all necessary information is present, the documents are approved by the Registrar of titles. After approval, the documents are signed by the land officer or legal representative, confirming the transaction's validity. The signed documents are then stored in physical files, often in cabinets or archives, to be retrieved when need occurs. The retrieval of this information requires individuals to physically visit the land offices, submit a request for the records, and wait for the staff to search the archives.

Although government efforts are aimed at digitizing the property records through the existing digital systems, like the National Land Information System (NLIS), much of the data is still stored manually and the migration to a digital system is ongoing.

4.2 STRENGTH OF THE CURRENT SYSTEM.

The current manual property records system in Uganda has well-established procedures for property transactions that have been utilized for years. These procedures provide a clear framework for property registration, transfers, and ownership verification, ensuring that users understand the necessary steps involved in these processes.

Many Ugandans, especially in rural areas, are familiar with the manual processes. This community engagement encourages face-to-face interactions between property owners and land officers, which can help build trust and provide opportunities for local authorities to educate residents about their rights and responsibilities regarding land as well as aid in making informed decisions regarding property ownership and disputes.

Another strength lies in the accessibility of this manual system by property owners that are less experienced with digital systems. In many rural communities, access to digital tools and the internet is limited, making the manual process more user-friendly for residents who may not possess advanced technical skills. This ensures that property owners can still navigate the system without needing specialized knowledge.

More-so, the Ugandan government's ongoing commitment to modernizing the land registration system through the National Land Information System (NLIS) reflects a

recognition of the importance of property recording for economic development of the country at large.

4.3 WEAKNESS OF THE CURRENT SYSEM.

Despite its strengths, the current manual property record system in Uganda is burdened by several critical weaknesses that severely limit its effectiveness and efficiency.

One of the most pressing issues is the fragmentation of data. Property records are maintained independently by various offices at the district level, creating a lack of coordination and leading to inconsistencies in record-keeping. This decentralized approach makes it difficult for property owners and legal representatives to access accurate and timely information about property ownership, complicating transactions and increasing the potential for disputes.

Furthermore, many records are outdated, poorly maintained, and fragile. This degradation poses a substantial obstacle to the transition toward a digital system, as migrating such data is both complex and resource-intensive.

Consequently, the current reliance of the system on paper records not only slows down the modernization process but also heightens the risk of errors during data entry and record-keeping. These human errors can lead to significant gaps in land ownership records, further complicating matters for property owners who seek to verify their ownership or resolve disputes.

Access to property records is also limited and cumbersome. The manual nature of the system necessitates physical visits to land offices, where retrieval times can be slow. The inconvenience of having to navigate bureaucratic hurdles often discourages individuals from pursuing property transactions, resulting in missed opportunities and financial losses.

In addition, there is a noticeable focus on land records while other family property records, such as commercial and residential properties, are often neglected. This narrow scope means that the system does not provide a comprehensive view of all types of family property holdings, limiting its overall effectiveness in managing diverse assets.

The vulnerability of physical records poses a significant risk to the integrity of property information. These records can be easily damaged by environmental factors or lost due to misplacement, which could lead to the permanent loss of vital information.

4.4 COMPARATIVE ANALYSIS OF THE STRENGTHS AND WEAKNESSES.

The current property record system in Uganda presents a mix of strengths and weaknesses that significantly impact its functionality and effectiveness.

A significant strength of the current system is the Ugandan government's commitment to modernizing land administration through initiatives like the Land Information System (LIS) and the National Land Information System (NLIS). Established procedures for property

transactions, although manual, offer users a clear pathway to navigate the system, particularly in rural areas where familiarity with these processes remains strong.

Despite its strengths, the system is plagued by several weaknesses. Data fragmentation is a major concern, as property records are maintained independently across various district offices, leading to inconsistencies and difficulties in retrieving accurate information. The reliance on manual data entry further exacerbates the problem, introducing human errors that can result in disputes and delays in property transactions.

Transitioning to digital systems presents significant opportunities for enhancing property record management. A centralized digital system could streamline transactions, reduce disputes, and improve data accuracy by consolidating information into a single, accessible platform that will encourage people to easily keep and manage their properties.

However, the system faces threats that could hinder progress. The vulnerability of physical records to loss or tampering poses a critical risk, as this could undermine the integrity of property information and hinder the government's modernization efforts to a digital system.

In evaluating the current property record system in Uganda, the weaknesses significantly outweigh the strengths. Therefore, addressing these critical weaknesses is essential for establishing a more reliable and efficient family property record management system.

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4.5 REQUIREMENTS OF THE NEW SYSTEM.

These are the requirements of the new system, A Family Property Records System.

4.5.1. USER REQUIREMENTS

The new system is easy to use with clear instructions and tooltips for navigating the system for the users. This was made possible by the use of the MySQL database with the users table in system development.

Users need a desktop or laptop to efficiently use the new system, which has simple forms for easy data entry, retrieval, and updating of property records data.

There is also be of user instructions that will support users on how to navigate the system. Users will be able to provide feedback or report issues from the data collection process easily via electronic platforms like e-mail. There will be assurance that user data and property records are kept confidential and safe from public.

4.5.2 SYSTEM REQUIREMENTS

In developing the new system, a combination of programming languages and technologies is utilized to ensure efficient functionality and user experience. PHP (Hypertext Preprocessor) serves as the primary server-side scripting language, enabling dynamic web page generation and handling form submissions to facilitate user interactions with the system. MySQL is employed as the database management system, providing a reliable platform for storing and retrieving property records, user information, and transaction data. The front-end interface is built using HTML (Hypertext Markup Language) for structuring web content, while CSS (Cascading Style Sheets) is used for styling and enhancing the visual presentation.

Additionally, JavaScript is incorporated to add interactivity to the user interface, allowing features such as real-time form validation and dynamic content updates.

Together, these programming languages and technologies create a robust and user-friendly system that effectively manages property records.

4.5.3 FUNCTIONAL REQUIREMENTS

The new features user authentication, allowing individuals to create accounts and securely log in with unique usernames and passwords. Users can manage profiles that contain personal information and contact details.

The system enables the entry of new family records, such as property details, various property types like residential, commercial and necessary documents, while providing efficient data retrieval capabilities. Additionally, users can upload and store supporting documents related to their family records for easy access.

4.5.4 NON-FUNCTIONAL REQUIREMENTS

The response time for data queries and actions in the new system is fast and the system is designed to accommodate up to 100 users without degradation of system performance.

There are secure login provisions into the database to allow only authorized access to the data collected hence securing data uploaded. There is a consistent user interface design with clear navigation path on how to navigate the new system.

The system is designed for maintainability, with well-documented code that allows for efficient bug fixes and updates.

4.6 CHAPTER SUMMARY.

The system study, analysis, and requirements elicitation process has provided valuable insights into the current functionality of the Family Property Records System and identified key areas for enhancement in the current manual recording system.

CHAPTER FIVE: SYSTEM DESIGN AND IMPLEMENTATION.

5.0 INTRODUCTION

This system focuses on the system design, implementation, testing and validation of the Family Property Records System.

5.1 SYSTEM DESIGN.

System design and modelling using Entity-Relationship Diagrams (E-R Diagram) provides a visual representation of the data model and that ensures a clear understanding of how data entities interact within the system.

5.1.1 ENTITY-RELATIONSHIP DIAGRAM

The Entity-Relationship Diagram is a key tool used to model the system's data structure. It identifies the entities within the system and defines the relationships between them. Below is an overview of the primary entities and relationships in the Family Property Record System:

An **entity** is a distinct object or concept within a system that has a set of attributes and is represented as a record in a database. Each entity instance is uniquely identifiable by a primary key. Entities can represent things like people, places, objects, or events that are relevant to the database. These are the entities for this system:

- **User:** Represents individuals who interact with the system, including roles and permissions.
- **Property:** Details about the property, such as name, location, type, and ownership.
- **Document:** Digital copies of relevant property documents.

A relationship is a connection or association between two or more entities that describes how they interact with each other. It defines how data in one entity relates to data in another entity, often through foreign keys that reference primary keys in other tables.

- A **User** can be associated with multiple **Properties** and have specific roles (e.g., owner, administrator).
- A **Property** can have one associated **Document**.
- **Document** is linked to specific **Property**.

This below is the system's E-R diagram

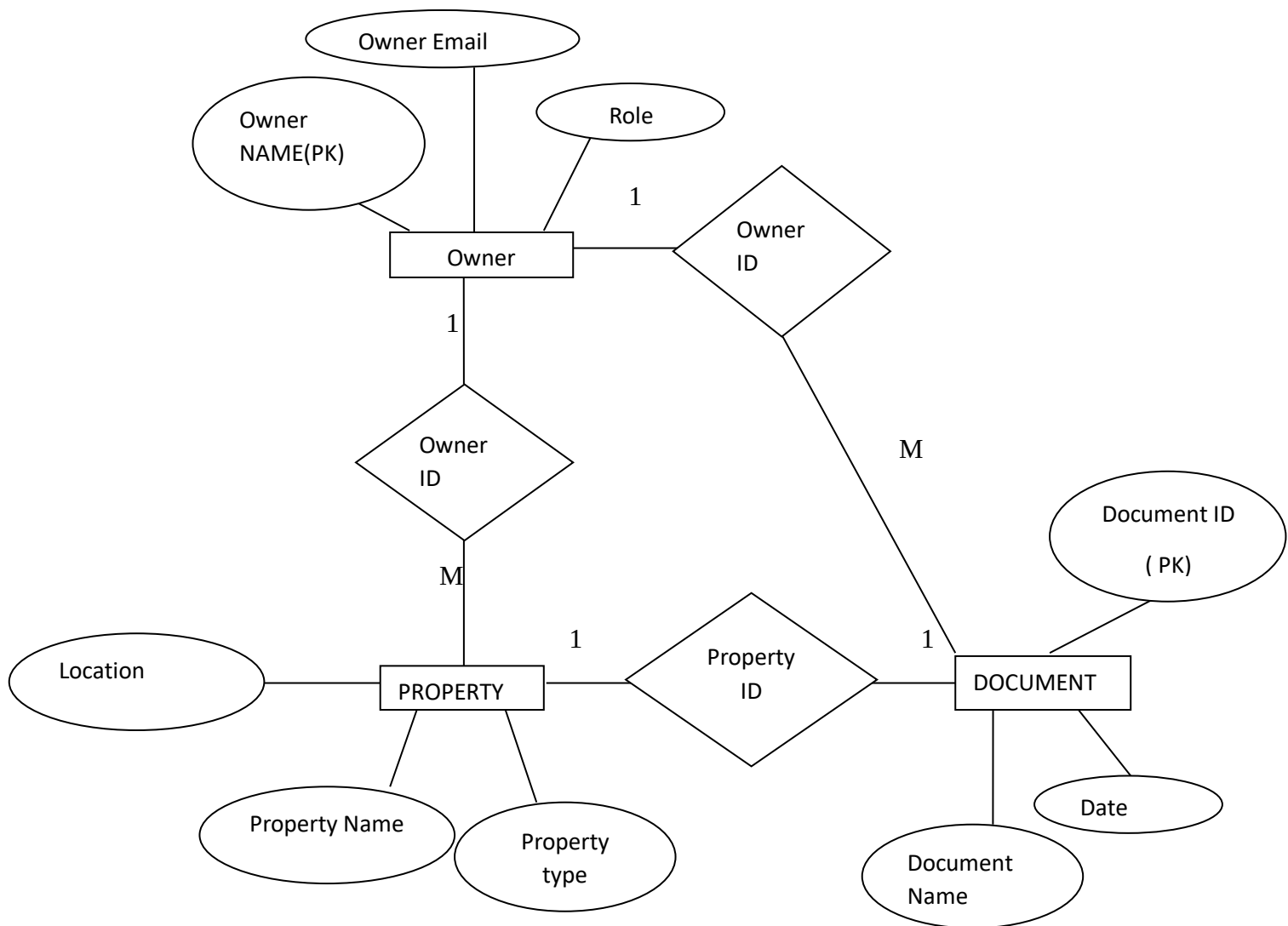
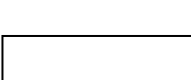


Figure 2: System E-R diagram

KEY

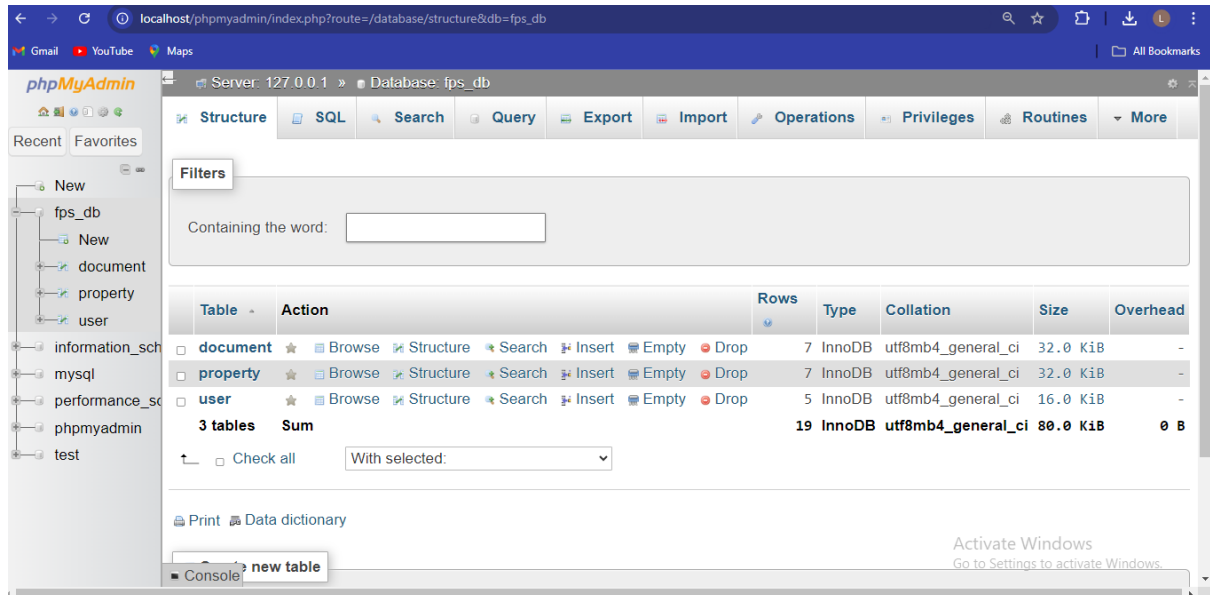
- PK- Primary Key
- 1-one
- M-many
-  -attribute
-  -relationship
-  - entity

5.2 SYSTEM LAYOUT

5.2.1 DATABASE LAYOUT

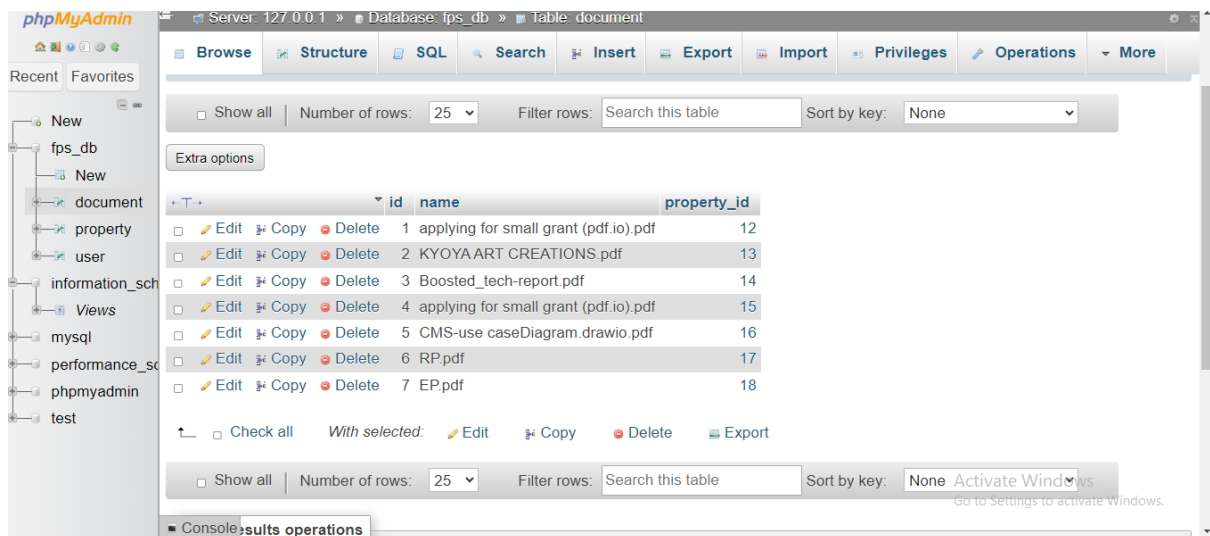
The system database is fps_db which stands for family property system_database.

This is the database interface;



The database has three tables and that is

-Document table for storing family documents;



The Document table is designed to store information about various documents related to properties within the system. Each entry includes essential fields such as id, which uniquely identifies each document, along with attributes like property_id to link the document to a specific property, and document_name.

-Property table for storing the property information;

The screenshot shows the phpMyAdmin interface for the 'property' table in the 'fps_db' database. The table has 8 columns: id, name, type, location, owner_id, last_edited_by, and last_edited_at. The data is as follows:

	id	name	type	location	owner_id	last_edited_by	last_edited_at
☐ Edit ☐ Copy ☐ Delete	12	Robert's Investments	Residential	Kyaliwajjala	3	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	13	Majanjo Investment	Industrial	Namugongo	3	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	14	Aine Dixon's Land	Land	kamuli	3	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	15	Robert's Land	Residential	Kamokya	1	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	16	house plot	Industrial	kamuli	1	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	17	risk_management_policy	Commercial	Banda	5	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	18	Policy	Industrial	Banda	6	NULL	NULL

The Property table serves as the core repository for all property-related information in the system. It includes a unique property_id for each property, along with fields like property_name, property_type, and location to provide essential details about the property. Additionally, the table may contain fields for date_added for tracking when the property was registered, and last_edited to monitor any updates made to the property records. This table is fundamental for managing property listings and facilitating efficient retrieval of property information by users.

-The user table for storing the system users;

The screenshot shows the phpMyAdmin interface for the 'user' table in the 'fps_db' database. The table has 5 columns: id, username, email, password, and role. The data is as follows:

	id	username	email	password	role
☐ Edit ☐ Copy ☐ Delete	1	Robert Bob	rob@gmail.com	\$2y\$10\$PZf25KnYJF8Ddc/ledw.Xe0yn8LRtmZLLE4yOYQewfm...	NULL
☐ Edit ☐ Copy ☐ Delete	2	root	root@gmail.com	\$2y\$10\$DG7/ippj0iPWPo3rKuXtMelbAwJ.Sn/tdcX/jv1q9FX...	1
☐ Edit ☐ Copy ☐ Delete	3	admin	admin@gmail.com	\$2y\$10\$075z1VIUW4o8yceBzFXsMuNcmCY0sRkzP9fZ89K3I/A...	1
☐ Edit ☐ Copy ☐ Delete	5	Lunah Lunah	lunahamanda1@gmail.com	\$2y\$10\$OMhd4/bnOXib.N6uKGEiquUDYW3sZj7OGqqz72Ct/X...	NULL
☐ Edit ☐ Copy ☐ Delete	6	One	lunahlunah50@gmail.com	\$2y\$10\$wMvMd51.mEzYOJRwSZopdu36M07w1.L0hX1gZgeyrfx...	NULL

The user table holds vital information about individuals who access the system. Each user is assigned a unique user_id for identification, along with key details such as username, email, and password to ensure secure authentication. The structure of this table is crucial for managing user access and permissions, providing a foundation for a secure and organized user experience within the system.

5.2.2 SYSTEM INTERFACE AND SAMPLE CODES.

These are the steps of navigating the Family Property Records System that is meant to store and manage family property documents;

STEP 1: -Access the Login Page.

Open your web browser and go to the login URL (<http://localhost/fps/index.php>) where you will click the register link and register your credentials as a new user and later log in with those same credentials on the login page.

This is the sample code for the register.php for the system;

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="assets/styles/style.css?v1" >
  <title>Register</title>
</head>
<body>
  <div class="container">
    <h1 class="heading-1">Family Property Keeper</h1>
    <div class="box form-box">
      <?php
        include("php/config.php");

        if(isset($_POST['submit']))
        {
          $username = $_POST['username'];
          $email = $_POST['email'];
          $password = $_POST['password'];
```

```

        $hashed_password
password_hash($password,PASSWORD_DEFAULT);

        //verifying the unique email
        $verify_query = mysqli_query($con, "SELECT email FROM
user WHERE email = '$email'");
        if(mysqli_num_rows($verify_query) != 0)
        {
            echo "<div class= 'message'>
                <p>This email is used, Try another One Please!</p>
                </div> <br>";
            echo " <a href='javascript:self.history.back()'><button
class='btn'>Go Back</button></a>";
        }
        else
        {
            mysqli_query($con, "INSERT INTO user(username, email,
Password) VALUES('$username', '$email', '$hashed_password')") or
die("Error Occurred");
            echo "<div class= 'message'>
                <p>Registered successfully!</p>
                </div> <br>";
            echo " <a href='index.php'><button class='btn'>Login
Now</button></a>";
        }

    }
    else
    {

        ?>
        <header>Register</header>
        <form action="" method="post">
            <div class="field input">
                <label for="username">Your Name</label>
                <input type="text" name="username" id="username"
autocomplete="off" required>
            </div>
            <div class="field input">
                <label for="email">Email</label>
                <input type="email" name="email" id="email"
autocomplete="off" required>
            </div>
            <div class="field input">
                <label for="password ">Password</label>

```

```

        <input type="password" name="password" id="password"
autocomplete="off" required>
    </div>
    <div class="field">
        <input type="submit" class="btn" name="submit"
value="Register" required>
    </div>
    <div class="links">
        Have an account already? <a href="index.php">Login In</a>
    </div>
</form>
</div>
<?php
    }
?>
</div>

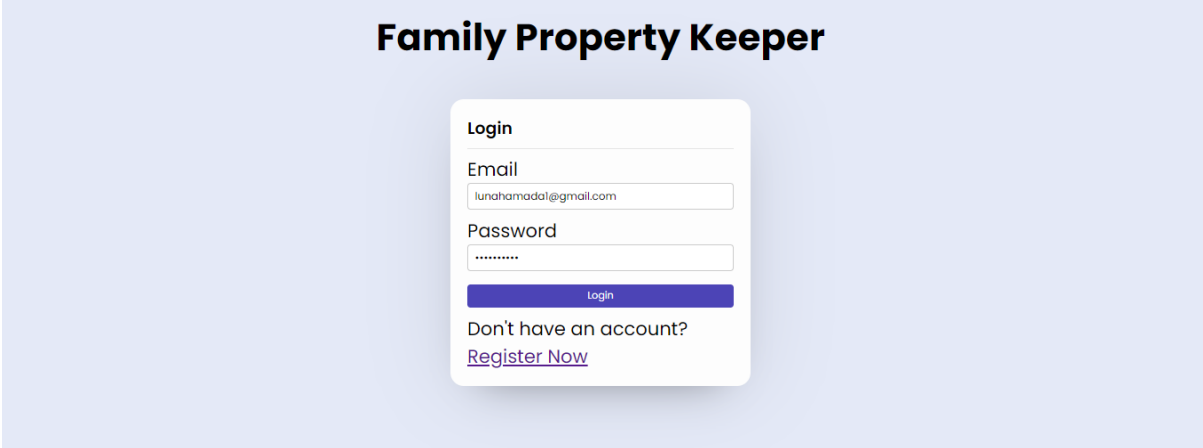
</body>
</html>

```

-Enter Credentials

Input your username and password in the respective fields.

Submit Login Form



The image shows a login form titled "Family Property Keeper". The form is centered on a light blue background. It contains a "Login" header, an "Email" field with the placeholder "lunahamadal@gmail.com", a "Password" field with masked characters "*****", a blue "Login" button, and a link "Don't have an account? Register Now".

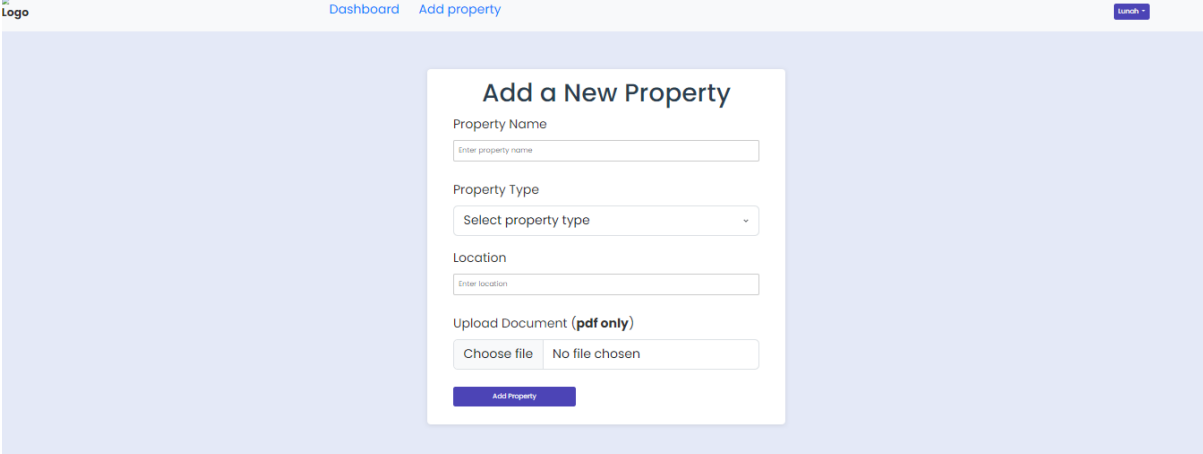
-Click the Login button to authenticate your credentials.

STEP 2: -Dashboard Navigation

Once logged in, you will be directed to the main dashboard, where you can view properties and access other features. Then you will be able to perform Actions;

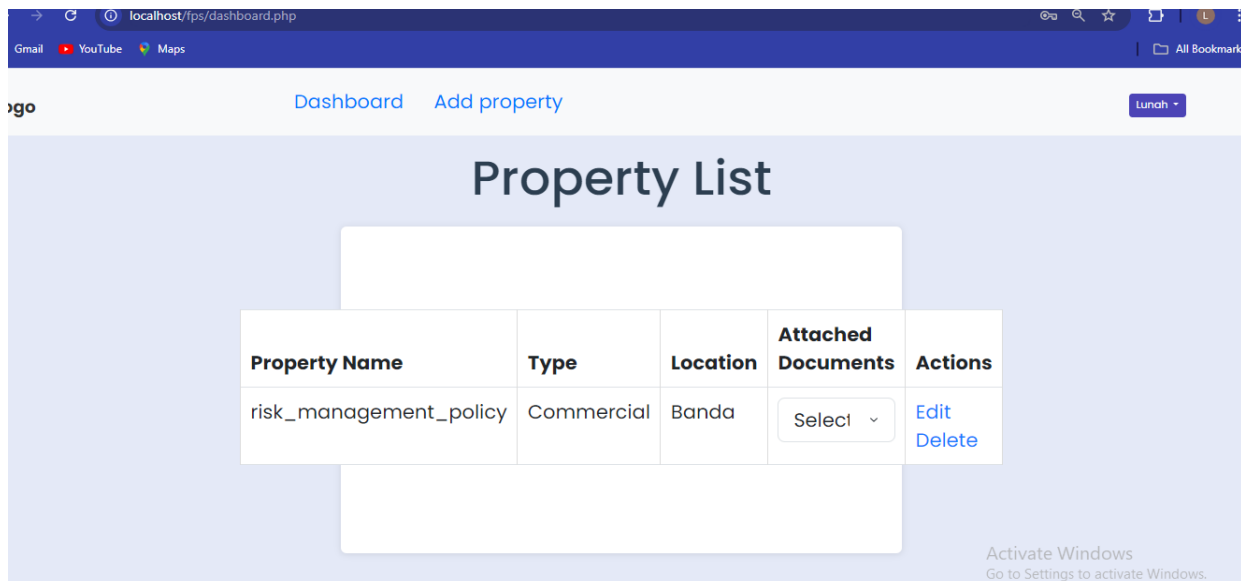
-Add new properties.

Here, you will press the add property button on the dashboard which you will click to add your document and its credentials in the system and upload it by clicking the add property button.



The image shows a dashboard with a navigation bar containing "Logo", "Dashboard", "Add property", and a user profile "Lunah". The main content area features a form titled "Add a New Property". The form includes fields for "Property Name" (placeholder: "Enter property name"), "Property Type" (a dropdown menu with "Select property type"), "Location" (placeholder: "Enter location"), and "Upload Document (pdf only)" (with "Choose file" and "No file chosen" buttons). A blue "Add Property" button is at the bottom of the form.

You can now view your documents in the dashboard.



One can only view his and only his documents while the admin can view all the documents.

This is the code for the add_property.php;

```
<?php
    session_start();

    include("php/config.php");
    if(!isset($_SESSION['valid']))
    {
        header("Location: index.php");
    }

?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard | Add property</title>
    <?php require_once "../parts/head_assets.php" ?>
</head>
<body>
    <?php require_once "../parts/nav.php" ?>
    <div class="container mt-5">
<?php
    include("php/config.php");

    if(isset($_POST['submit']))
    {
        $p_name = $_POST['name']; // must be unique in the database
```

```

$sp_location = $_POST['location'];
$sp_type = $_POST['type'];
$owner_id = (int)$_SESSION['id'];

$query = "INSERT INTO property(`name`, `type`, `location`,
`owner_id`) VALUES (?, ?, ?, ?)";
// Prepare the SQL statement
$stmt = mysqli_prepare($con, $query);
// Check if preparation was successful
if ($stmt === false) {
    die("MySQL prepare statement failed: " . mysqli_error($con));
}
mysqli_stmt_bind_param($stmt, 'sssi', $p_name, $p_type, $p_location,
$owner_id);
// 'ssss' means 4 strings; you can use 'i' for integers (in this case, the
owner_id is an integer)

if (!mysqli_stmt_execute($stmt)) {
    die("MySQL execute statement failed: " . mysqli_error($con));
}
mysqli_stmt_close($stmt);

// document upload
$targetDir = "documents/";
// echo $_FILES['doc']['name'];
// echo $_FILES['doc']['error'];
// exit();
if (isset($_FILES['doc']) && $_FILES['doc']['error'] ==
UPLOAD_ERR_OK)
{
    // Extract file information
    $fileTmpPath = $_FILES['doc']['tmp_name'];
    $fileName = basename($_FILES['doc']['name']);
    $fileType = $_FILES['doc']['type'];
    $fileExtension = strtolower(pathinfo($fileName,
PATHINFO_EXTENSION));

    // check if the file is a pdf
    if ($fileExtension == "pdf")
    {
        $p_name = mysqli_real_escape_string($con, $p_name);

        $result = mysqli_query($con, "SELECT id FROM property
WHERE name = '$p_name'") or die("Cant find owner id");
        $num_rows = mysqli_num_rows($result);
        // echo $num_rows;
    }
}

```

```

        if ($num_rows == 1)
        {
            $row = mysqli_fetch_assoc($result);
            $propertyId = (string)$row['id'];
        }
        else{
            die("Not equal to one row selected");
        }
        $newFileName = $propertyId . '-' . $fileName;

        $targetFilePath = $targetDir . $newFileName;

        // Move the uploaded file to the target directory
        if (move_uploaded_file($fileTmpPath, $targetFilePath)) {
            $parts = explode('-', $newFileName, 2);
            $saved_doc = $parts[1];
            $doc_query = "INSERT INTO document(name, property_id)
VALUES('$saved_doc', '$propertyId')";
            mysqli_query($con, $doc_query) or die("Failed to save document
details");
            echo "The file has been uploaded and saved as: " . $saved_doc;

        } else {
            die("Error moving the uploaded file.");
        }

    }
    else
    {
        die(" Only PDF files Allowed");
    }
}
else
{
    exit("Error: Document was not uploaded!");
}

echo "<div class= 'message'>
<p>Property added successfully!</p>
</div> <br>";
        echo " <a href='dashboard.php'><button class='btn'>Back to
Home</button></a>";
    }
    else
    {
        ?>

```

```

<h2 class="text-center">Add a New Property</h2>
<form action="" method="POST" enctype="multipart/form-data">

    <!-- Property Name -->
    <div class="mb-3">
        <label for="propertyName" class="form-label">Property
Name</label>
        <input type="text" name="name" class="form-control"
id="propertyName" placeholder="Enter property name" required>
    </div>

    <!-- Property Type Selection -->
    <div class="mb-3">
        <label for="propertyType" class="form-label">Property Type</label>
        <select class="form-select" id="propertyType" name="type"
required>
            <option selected disabled>Select property type</option>
            <option value="Residential">Residential</option>
            <option value="Commercial">Commercial</option>
            <option value="Industrial">Industrial</option>
            <option value="Land">Land</option>
        </select>
    </div>

    <!-- Location -->
    <div class="mb-3">
        <label for="locationName" class="form-label">Location</label>
        <input type="text" name="location" class="form-control"
id="locationName" placeholder="Enter location" required>
    </div>

    <!-- Upload Document -->
    <div class="mb-3">
        <label for="uploadDocument" class="form-label">Upload Document
(<strong>pdf only</strong>)</label>
        <input class="form-control" name="doc" type="file" accept=".pdf"
id="uploadDocument" required>
    </div>

    <!-- Submit Button -->
    <button type="submit" name="submit" class="btn btn-
primary">Add Property</button>

</form>

<?php } ?>

```

```

        <?php require_once "../parts/bottom_assets.php" ?>
    </div>
</body>
</html>

```

STEP 3: Log Out

When finished, click the dropdown button in the right upper corner under your username to logout.



You will be redirected to the login page, confirming successful logout.

5.3 DATA DEFINITIONS.

These are the definition of terms or data used in the new system;

Family: This is a group of individuals related by blood, marriage, or legal ties, sharing a common household and mutual responsibilities. In this system, it refers to those with legal interests in property ownership and management.

Family Records: These are documented information related to a family's property ownership and legal rights, including title deeds, wills, and vital records such as birth and marriage certificates. These records are essential for accurate property management.

System: This is a structured digital platform that enables users to store, manage, and access family property records. It consists of hardware, software, and databases designed to streamline property management processes.

User: This is an individual who interacts with the Family Property Record System to access or manage property-related information, including family members, legal representatives, and government officials. The user can have different roles like admin, user and guest.

User Authentication: This is the process of verifying a user's identity before granting access to the Family Property Record System, typically through unique credentials like a username and password. This ensures that only authorized individuals can access sensitive information.

5.4 SYSTEM IMPLEMENTATION.

This involves the practical construction of the system. This stage includes coding, integrating various components, and deploying the system in a test environment. It requires careful attention to detail and adherence to best practices to ensure that the system is efficient, and scalable. These strategies were employed;

Development Environment Setup; Availability of required hardware resources like laptops, phones, notebooks and software resources like Visual Studio Code software, MySQL relational database, xampp in the system development process were ensured.

Coding and Development; Here, the actual coding using programming languages like PHP (Hyper-text Preprocessor), CSS (Cascading Style Sheets) and HTML (HyperText Markup Language) and connection of the system to the database will be done. User interfaces were aligned with the user requirements and core modules based on the system specifications.

Integration of the system; Individual modules are combined into a cohesive system, ensuring seamless communication and data exchange between components. Data persistence mechanisms is implemented using a database management system (DBMS) MySQL, to store and retrieve information efficiently.

Deployment of the system; It will be deployed in a test environment that stimulates the production environment. This will prepare the system for potential challenges in transitioning from developing to production.

5.5 SYSTEM TESTING AND VALIDATION.

System testing is a critical step in ensuring the system's reliability, functionality, and performance. These testing strategies employed:

Individual components and functions are tested to verify their correctness and functionality, the combined modules are tested to ensure that they work well together with no issues and then the whole system is tested to ensure that it meets the system functional and non-functional requirements. The system performance is assessed under different conditions and workloads. Involvement of end-users in testing to confirm the system meets their needs and expectations is exercised. This series of testing will help the developer to identify and resolve interface related issues.

System validation ensures that the system meets user needs and expectations and is fit for its intended purpose. These strategies will be employed.

User Acceptance Testing (UAT) whereby actual users are involved in pilot testing the system was done. Documentation of the system development process whereby comprehensive user manuals and documentations to facilitate users to understand and adopt to the system were provided and made available for future use. Final Evaluation whereby the system is assessed in terms with the research goals and objectives to find out if it matches with them was done. The system is accurate.

5.6 CHAPTER SUMMARY.

In conclusion, this chapter has thoroughly examined the system design, implementation, testing, and validation of the Family Property Records System. The systematic approach taken throughout the project has been pivotal in addressing the complexities associated with managing family property records in Uganda. By focusing on robust design principles, we ensured that the system is not only functional but also user-friendly, enabling users to navigate through the various features with ease. The implementation phase demonstrated the practical application of theoretical frameworks, resulting in a cohesive and efficient digital platform tailored to meet the needs of its users. Rigorous

testing and validation processes confirmed the system's reliability, highlighting its ability to add and manage family property documents.

Some of the findings from the system implementation include; the Change profile button, edit and delete buttons on the dashboard were not functional.

This comprehensive development process not only enhances the management of family property records but also contributes to broader societal goals, such as promoting transparency, accessibility, and empowerment within communities. The insights gained from this chapter emphasize the significance of a well-structured system in facilitating better property management practices, ultimately paving the way for improved socio-economic stability for families across Uganda.

CHAPTER SIX: DISCUSSIONS, RECOMMENDATIONS AND CONCLUSIONS.

6.0. INTRODUCTION

This chapter presents the key conclusions drawn from the development and implementation of the Family Records System. It provides the relevant discussions, recommendations, areas for further research, limitations and conclusions of the study.

6.1. DISCUSSIONS.

The new Family Property Records System has brought many improvements compared to the old manual system. One of the biggest changes is the centralized database, which brings all property records of different property types like residential, commercial into one digital platform. This has helped fix issues with missing and inconsistent data that were hard to notice in the old system. Now, users can access accurate information more easily, solving the problems caused by scattered records in the manual system.

The system also offers real-time updates, which means property records are immediately updated when any changes are made. This is much faster than the old manual system, where records could take days or weeks to update, often leading to disputes. Real-time updates have made the process of checking property records more reliable and faster, making it easier to manage property transactions.

In this system, users can only update and view their property documents without the invasion of privacy from other family members and only the family head as the admin can view them. I discovered that in the long run, an addition of next of kin has to be added for the sustainability of the system in case the admin is no longer active as well as login credentials using national identification numbers should be used for more security.

Security is another key improvement. The new system uses encryption and secure logins to protect sensitive information, unlike the manual system, which was open to tampering, loss, or unauthorized access. The enhanced security features make sure that only authorized users can access property records, keeping them safe from misuse.

However, there are still some challenges, especially for people who are not familiar with using digital tools. The system requires users to have basic digital skills like logging in, signing up and operating a system interface, which may be a problem for those in rural areas who are less used to technology. This shows a need for training programs to help people learn how to use the system.

During the migration of data from the old manual system to the new digital platform, several errors and missing information were uncovered. For example, some property records had duplicate entries, where the same property was listed under different owners, and key details like ownership transfer dates were missing. Additionally, there were incomplete property addresses and inconsistent or misspelled names of property owners. These issues highlight

the need for data cleaning and validation to ensure the accuracy of the records in the new system. Correcting these errors is essential to provide reliable property information and prevent future disputes.

Surprisingly, I discovered that mostly men participate in family property ownership, recording and management and the women are left out of the whole family property system which displays a high gender inequality in Ugandan families today recording property management and ownership which hinders our economic development as a country. This system will allow even women with their laptops to have their property details uploaded and viewed by them and admin, so their male family members have no access too denying and manipulating their property.

Overall, the new Family Property Records System has made big improvements in how various family property records are managed, providing better efficiency, security, and transparency. People can now stop recording their properties manually and start uploading and managing them using this digital system. However, the system still faces challenges with user adaptation and the quality of old data being transferred to the new platform that need to be addressed soon not later.

6.2. RECOMMENDATIONS.

Based on my research and development of the Family Property Record System, I strongly recommended that this system be adopted by various groups, including individuals, families, government agencies, legal representatives, and community organizations involved in family property management. The new system can greatly improve how their various property records are managed.

Government agencies responsible for land registration should use the Family Property Record System to streamline their operations. By digitizing records, these offices can reduce the time it takes to process transactions and minimize errors. This will allow them to provide better service to citizens who need information about family properties.

Legal representatives, such as lawyers and real estate professionals, will also benefit from the system. They can access up-to-date property records and generate reports on ownership history, which will help them give better advice to their clients and handle property transactions more effectively.

Additionally, community organizations, including non-governmental organizations (NGOs) focused on housing and land rights, should use the Family Property Record System to educate community members about their property rights. By providing access to digital records, these organizations can empower families to understand their ownership rights and navigate property transactions more easily.

Families and individuals are also encouraged to use this system to manage their property records effectively. The system allows them to easily access information, verify ownership, and track important documents and transactions. By adopting this digital platform, families can enhance their understanding of property rights and make informed decisions regarding their assets.

More-so, educational institutions that teach land management, real estate, and law can use the system as a practical learning tool. Students will gain hands-on experience with property management and data handling, preparing them for future careers in these fields.

To ensure the successful use of the Family Property Record System, it is important to provide training for all users. Training programs should be designed for different groups, such as government officials, legal representatives, and community members. These sessions should cover how to navigate the system, enter data correctly, and access property records. Hands-on workshops and online resources can help users learn effectively and gain confidence in using the system. Additionally, ongoing support and refresher courses should be available to help users keep up with any updates or new features, ensuring everyone feels comfortable using the digital platform.

In summary, adopting the Family Property Record System will bring significant improvements to property management in Uganda. It will help stakeholders work more efficiently, access accurate information, and promote transparency. This system represents an important step forward in modernizing land administration and improving the management of family property records.

6.3. LIMITATIONS OF THE STUDY.

The research study on the Family Property Record System faced several limitations that affected its overall scope and effectiveness. One major challenge encountered was related to data collection. During the collection of questionnaires and interviews, some individuals were reluctant to provide information due to concerns about privacy and data security even with the implementation of the study's ethical considerations. This reluctance limited the depth of insights gathered, as not all stakeholders felt comfortable sharing their experiences and opinions regarding the existing property management practices.

Additionally, logistical issues posed challenges in the research process. I encountered difficulties in transportation due to limited funds and bad roads, which made it harder to reach certain areas, particularly rural communities. This limited access restricted the ability to engage a broader range of participants, thereby affecting the representativeness of the sample used for the study.

Moreover, there were financial constraints that impacted the research. The lack of sufficient funds hindered the ability to collect data from some participants who required more monetary allowance. This limitation made it difficult to employ advanced data collection methods that could have enhanced the quality of the findings.

The study was faced by the difficulty in reaching out to government officials during the data collection phase. Many officials were often unavailable for interviews due to busy schedules and conflicting responsibilities, which hindered access to valuable insights about existing property management practices and policies. This challenge limited the depth of information gathered from a crucial stakeholder group, impacting the overall comprehensiveness of the research findings.

Unfortunately, the timeline for the research was constrained, which limited the ability to conduct follow-up interviews or gather additional data after the initial collection phase as well as develop the system to the study's satisfaction. This lack of follow-up reduced the opportunity to clarify responses or explore emerging themes in greater detail.

Some of the participants exhibited hostile and poor attitudes towards this study, which made the interview process more challenging. This negative response from a few individuals created an uncomfortable atmosphere, potentially affecting the quality of information obtained.

Overall, these limitations highlight the challenges faced during the research study and suggest areas for improvement in future investigations.

6.4. AREA FOR FURTHER RESEARCH.

Future research on the Family Property Record System can explore several important areas to enhance the effectiveness and usability of the system. One potential area for further study is the impact of digital literacy on user adoption. Understanding the barriers that individuals face in using the system, especially in rural communities, can inform the development of targeted training programs. Researching how to improve digital skills among users will be crucial for maximizing the benefits of the new system.

Another important area for further research is the integration of the Family Property Record System with other government databases. Exploring how the system can interact with existing records from agencies such as tax authorities or land registries could provide a more comprehensive view of property ownership. This integration would facilitate better data sharing and enhance the accuracy of property records.

The implementation of the system on mobile platforms could be explored to enhance accessibility and user engagement. By developing a mobile application, users would have easier access to key features and real-time updates, enabling broader reach and more convenience. Additionally, mobile integration would allow the system to utilize mobile-specific functionalities such as push notifications and GPS, making the system more interactive and user-friendly.

Additionally, studies could investigate the long-term effects of the system on property management. Researching how the new system influences property disputes, transaction speeds, and user satisfaction over time will help evaluate its success and identify areas for improvement. Gathering feedback from users after extended use can provide valuable insights into the system's strengths and weaknesses.

A critical area of study would be examining the role of women in property recording. Since I realized that men were more frequently involved in the property management process, it is important to explore how women engage with the Family Property Record System. Understanding women's experiences, challenges, and perspectives regarding property ownership and recording can lead to more inclusive practices and policies that empower female property owners. This research could also identify barriers that women face in accessing and utilizing the system effectively.

I believe that pursuing these areas for further research will not only strengthen the Family Property Record System but also contribute to the overall field of property management and digital record-keeping while ensuring that the needs of all users, particularly women, are adequately addressed.

6.5. CONCLUSION

This research has thoroughly investigated the challenges associated with family property recording and management in Uganda, revealing critical issues stemming from fragmented and outdated record-keeping practices. The study highlights significant potential for social problems, including family disputes over inheritance and property rights, which can escalate into severe conflicts, including loss of property and even violence. The lack of formal documentation among many landowners further exacerbates the risk of legal and financial insecurity, paving the way for property fraud.

The primary objective of this study was to develop a centralized Digital Family Property Records System that efficiently stores and manages family property documents while reducing fraud risk. The research successfully evaluated existing property recording methods, highlighting the inadequacies of manual systems in addressing the complexities of family dynamics and cultural beliefs that influence property management in Uganda.

Key findings indicate that traditional recording methods often lead to inconsistencies and disputes due to poor record-keeping practices and a lack of awareness of legal rights among family members. Furthermore, the study identified essential features for the proposed system, including secure access, automated data entry, and user-friendly interfaces designed to accommodate users.

The development of the Family Property Record System represents a vital step toward modernizing property management in Uganda. By providing a secure and centralized platform for family records, the system can significantly reduce the inefficiencies associated with fragmented documentation and improve the overall accuracy and accessibility of property information. Implementing this system fosters greater transparency, reduces disputes in the long run, and empower families to make informed decisions regarding their property, ultimately contributing to a more stable and equitable land management framework in Uganda. I can gladly say that the research study was mostly successful!

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APPENDICES

Appendix I: EXPLANATORY NOTES

The research involved visits to several key locations to gather data and insights from various stakeholders. Kampala, the capital city, served as a central hub for conducting interviews and surveys with participants. Insights gained from stakeholders in Kampala were essential for understanding the regulatory framework surrounding property ownership and documentation.

The methodology employed in this study was a mixed-methods approach, combining qualitative data from in-depth interviews (IDIs) with quantitative data collected through surveys. This approach enabled a comprehensive understanding of the cultural, social, and legal factors that influence property management in Uganda. By integrating perspectives from various stakeholders, the research aimed to develop a robust system that meets the needs of families.

Key locations visited during the research included family homes. Overall, the visits and interactions with various stakeholders were crucial for gathering relevant data and understanding community needs. This comprehensive approach ensured that the developed Family Property Record System is aligned with the realities faced by families in Uganda, ultimately contributing to better property management and dispute resolution.

Appendix II: CONSENT FORM AND QUESTIONAIRE FOR PARTICIPATION IN RESEARCH STUDY.

CONSENT FORM FOR PARTICIPATION IN RESEARCH STUDY

Title of Study: A Family Property Record System

Area: Kampala, Uganda

Principal Researcher: Kobusingye Muramuzi Lunah

Contact Information: lunahamanda@gmail.com

Introduction:

You are invited to participate in a research study conducted by Kobusingye Muramuzi Lunah from Kyambogo University. Before you decide to participate, it is important to understand the purpose of the study, the procedures involved, and your rights as a participant. Please read the following information carefully and feel free to ask any questions.

Purpose of the Study:

The study aims to assess and document current methods, identify difficulties, and understand the perceptions of individuals regarding property recordkeeping. By gathering insights and feedback, the research seeks to inform the design of a tailored family property record system that is efficient, accessible, and meets the specific needs

of Ugandan families to develop a centralized, secure, efficient and user-friendly digital family records system that stores family property related documents and reduces the risk of property fraud of these properties.

Procedures:

If you agree to participate in this study, you will be asked to complete a questionnaire. The questionnaire will take approximately 10-15 minutes to complete and will include questions about your property ownership, recordkeeping practices, and your views on developing a property record system.

Voluntary Participation:

Your participation in this study is entirely voluntary. You have the right to withdraw from the study at any time without any consequences.

1

QUESTIONNAIRE FOR THE RESEARCH STUDY ON FAMILY PROPERTY RECORD SYSEM.

Personal and Demographic Information

- Can you please introduce yourself and tell me a bit about your background?(include your name, age, gender)

Ahmuza Branch, 33, female.

- What is your role in your family concerning property management?

Spouse

2. Property Ownership and Management

- What type of property do you own or manage?

Rental 00

- How did your family acquire this property (e.g., purchase, inheritance, gift)?

Purchase.

- How do you and your family currently manage your property records?

Kept by lawyers

3. Record-Keeping Practices

- What types of records do you keep for your family property (e.g., title deeds, purchase agreements)?

None

- How are these records stored, and who is responsible for maintaining them?

Manually, husband

- Have you encountered any challenges in keeping these records organized and secure?

Yes

4. Challenges and Issues

- Can you describe any specific difficulties you've faced regarding property ownership and record-keeping?

Loss of Records

- How do disputes or misunderstandings typically arise concerning family property?

Lack of formal documents.

- Have you ever lost important property documents? If so, how did you handle the situation?

Yes, went through Land Board to acquire another

5. Perceptions and Attitudes

- How important do you think it is to have a formal system for recording family property?

Very Important

- How do cultural beliefs influence property ownership and management in your family?

Men are the ones who own property

- Would you be open to adopting a new system for managing property records? Why or why not?

K [V^I-C/VY(cm

- Who is typically responsible for managing family property in your family?

Husband.

6. System Design and Features

- What features would you find most helpful in a family property record system (e.g., secure storage, easy access, update capabilities)?

Secure storage.

- How would you prefer to access your property records (e.g., mobile app, web platform)?

Mobile app.

How would you prefer to access your property records

(e.g., mobile app, web

- What concerns do you have about transitioning to a digital record-keeping system?

Operating the system

7. Additional Feedback

- What improvements do you think are necessary in the current property management practices in your community?

Accessing files

- Do you have any suggestions for features or functionalities in the proposed property record system?

Storage

- Is there anything else you would like to share about your experience with family property management?

THANK YOU FOR YOUR PARTICIPATION, YOUR RESPONSE IS CONFIDENTIAL AND SECURE.

Confidentiality:

All information collected in this study will be kept confidential. Your responses will be anonymized,.

Benefits and Risks:

There are no direct benefits or risks associated with participating in this study but discussions about family property issues could trigger emotional responses or distress. However, your participation will contribute to the development of a system that can potentially benefit families in managing their property records more effectively.

Contact Information:

If you have any questions or concerns about this study, you can contact the principal researcher, Kobusingye Muramuzi Lunah at lunahamandal@gmail.com or 0787147609.

Consent:

By signing this form, you acknowledge that you have read and understood the information provided above and agree to participate in this study.

Participant's Name: Ahumuza Brenda.

Participant's Signature: 
25/7/2024

information
provided
above and
agree to
participate
in this
study.

Date:

CONSENT FORM FOR PARTICIPATION IN RESEARCH STUDY

Title of Study: A Family Property Record System

Area: Kampala, Uganda

Principal Researcher: Kobusingye Muramuzi Lunah

Contact Information: lunahamanda@gmail.com

Introduction:

You are invited to participate in a research study conducted by Kobusingye Muramuzi Lunah from Kyambogo University. Before you decide to participate, it is important to understand the purpose of the study, the procedures involved, and your rights as a participant. Please read the following information carefully and feel free to ask any questions.

Purpose of the Study:

The study aims to assess and document current methods, identify difficulties, and understand the perceptions of individuals regarding property recordkeeping. By gathering insights and feedback, the research seeks to inform the design of a tailored family property record system that is efficient, accessible, and meets the specific needs of Ugandan families to develop a centralized, secure, efficient and user-friendly digital family records system that stores family property related documents and reduces the risk of property fraud of these properties.

Procedures:

If you agree to participate in this study, you will be asked to complete a questionnaire. The questionnaire will take approximately 10-15 minutes to complete and will include questions about your property ownership, recordkeeping practices, and your views on developing a property record system.

Voluntary Participation:

Your participation in this study is entirely voluntary. You have the right to withdraw from the study at any time without any consequences.

Confidentiality:

All information collected in this study will be kept confidential. Your responses will be anonymized,.

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Contact Information:

If you have any questions or concerns about this study, you can contact the principal researcher, Kobusingye Muramuzi Lunah at lunahamandal@gmail.com or 0787147609.

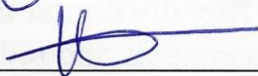
Consent:

By signing this form, you acknowledge that you have read and understood the information provided above and agree to participate in this study.

Participant's Name:

Kyobutungi Hellen

Participant's Signature:

ire: 

Date: ad-I

QUESTIONNAIRE FOR THE RESEARCH STUDY ON FAMILY PROPERTY RECORD SYSEM.

Personal and Demographic Information

- Can you please introduce yourself and tell me a bit about your background?(include your name, age, gender)

Am Kyobutung; Hellen, 25 years and a female

- What is your role in your family concerning property management?

I am a family member

2. Property Ownership and Management

- What types of family property do you own or manage?

I own a business

- How did your family acquire this property (e.g., purchase, inheritance, gift)?

It was purchased

e How do you and your family currently manage your property records?

_rQ-CTDD

3. Record-Keeping Practices

- What types of records do you keep for your family property (e.g., title deeds, purchase agreements)?

Purchase agreements

- How are these records stored, and who is responsible for maintaining them?

aru

- Have you encountered any challenges in keeping these records organized and secure?

No

4. Challenges and Issues

- o Can you describe any specific difficulties you've faced regarding property ownership and record-keeping?

Acquiring the necessary documents. He has a lot of bureaucratic tendencies

- How do disputes or misunderstandings typically arise concerning family property?

o V tJQY

mis comm

- o Have you ever lost important property documents? If so, how did you handle the situation? _____t

nCVQ-t

lc«-L

document

and

5. Perceptions
Attitudes

- How important do you think it is to have a formal system for recording family property?

It is important because information is

- How do cultural beliefs influence property ownership and management in your family?

Some cultures don't believe that a woman should work or have more income so women are not involved in property management

- Would you be open to adopting a new system for managing property records? Why or why not?

- Who is typically responsible for managing family property in your family?

—1 (TQx

6. System Design and Features

- What features would you find most helpful in a family property record system (e.g., secure storage, easy access, update capabilities)?

- How would you prefer to access your property records (e.g., mobile app, web platform)?

What concerns do you have about transitioning to a digital record-keeping system?

no

7. Additional Feedback

- What improvements do you think are necessary in the current property management practices in your community?

An IT system to manage records efficiently

- Do you have any suggestions for features or functionalities in the proposed property record system?

10 0

- Is there anything else you would like to share about your experience with family property management?

NO

THANK YOU FOR YOUR PARTICIPATION, YOUR RESPONSE IS CONFIDENTIAL AND SECURE.

Appendix III: INTERVIEW GUIDE FOR THE RESEARCH STUDY ON FAMILY PROPERTY RECORD SYSTEM.

(This is the consent form for the interviewee)

CONSENT FORM FOR PARTICIPATION IN RESEARCH STUDY

Title of Study: A Family Property Record System

Area: Kampala, Uganda

Principal Researcher: Kobusingye Muramuzi Lunah

Contact Information: lunahamanda@gmail.com

Introduction:

You are invited to participate in a research study conducted by **Kobusingye Muramuzi Lunah** from **Kyambogo University**. Before you decide to participate, it is important to understand the purpose of the study, the procedures involved, and your rights as a participant. Please read the following information carefully and feel free to ask any questions.

Purpose of the Study:

The study aims to assess and document current methods, identify difficulties, and understand the perceptions of individuals regarding property record-keeping. By gathering insights and feedback, the research seeks to inform the design of a tailored family property record system that is efficient, accessible, and meets the specific needs of Ugandan families to develop a centralized, secure, efficient and user-friendly digital family records system that stores family property related documents and reduces the risk of property fraud of these properties.

Procedures:

If you agree to participate in this study, you will be asked to complete a questionnaire. The questionnaire will take approximately 10-15 minutes to complete and will include questions about your property ownership, record-keeping practices, and your views on developing a property record system.

Voluntary Participation:

Your participation in this study is entirely voluntary. You have the right to withdraw from the study at any time without any consequences.

• What types of records do you keep for your family property (e.g., title deeds, purchase agreements)?

• How are these records stored, and who is responsible for maintaining them?

• Have you encountered any challenges in keeping these records organized and secure?

662 words English (Uganda)

Confidentiality:

All information collected in this study will be kept confidential. Your responses will be anonymized.

Benefits and Risks:

There are no direct benefits or risks associated with participating in this study but discussions about family property issues could trigger emotional responses or distress. However, your participation will contribute to the development of a system that can potentially benefit families in managing their property records more effectively.

Contact Information:

If you have any questions or concerns about this study, you can contact the principal researcher, **Kobusingye Muramuzi Lunah** at lunahamanda1@gmail.com or 0787147609.

Consent:

By signing this form, you acknowledge that you have read and understood the information provided above and agree to participate in this study.

Participant's Name: Bamy Kato

Participant's Signature: [Signature]

Date: 28/07/2024

Consent form of interviewee



WhatsApp Audio 2024-10-01 at 10.28.39.mp4

Audio of interview

Appendix IV: GOOGLE RESPONSE FORM LINK.

https://docs.google.com/forms/d/1tHy_EgZ-Y1h-LPmR4crht0qj8JvmwCjim28tGSnb2f0/edit (This is the google forms link)

https://docs.google.com/spreadsheets/d/1mV6vmfNSBj_qyZY4P2pT3XOyDQ5wZOS8wfdUvMOIoVs/edit?usp=sharing (This is the link for the response)

Appendix V: OBSERVATION CHECKLIST

Appendix V: OBSERVATION CHECKLIST

Project Title: A Family Property Record System in Uganda

Observer Name: Kobusingye Muramuzi Lunah

Location: Kampala, Uganda

☐ **Interaction Type:**

- Formal meetings
- ✓ Informal discussions ✓ family meeting
- Public consultations

☐ **Participants Involved:** [e.g., Families, Government officials]

☐ **Interaction Context:** [e.g., Office setting, Community meeting, home compound]

☐ **Key Points Discussed:** [e.g., Property boundaries, Ownership disputes]

☐ **Responses and Reactions:** [e.g., Satisfaction, Frustration, Joy, Anger]

Record Keeping Practices

- **Type of Records Observed:**
 - Title deeds
 - Wills
 - Rental contracts ✓
 - Trust documents
- **Storage Method:** [e.g., Physical filing, Digital database]
- **Record Maintenance:** [e.g., Frequency of updates, Condition of records] poor
- **Challenges Noted:** [e.g., Loss or damage, Unauthorized access]

Additional Notes

- **Unexpected Observations:** Women were not allowed to give view on property management
- **Potential Impact on Research:** Raised need for user authentication on digital property records system
- **Follow-up Actions Required:** Digital family property records system

Observer's Signature:  Kobusingye Muramuzi Lunah

Date: 30th July, 2024.

Researcher's Observation checklist

Appendix VI: RESEARCH BUDGET

RESEARCH BUDGET ESTIMATE FOR THE FAMILY PROPERTY RECORD SYSTEM PROJECT IN UGANDA SHILLINGS (UGX).

Item	Unit Quantity	Overall period	Unit Cost (UGX)	Overall Cost (UGX)	Description	Activity	Timeline
Personnel Costs	1 Head Researcher	4 months	2,000,000	8,000,000	Salary for 1 head researcher for 4 months	Hiring head researcher	July-August
	2 Research Assistant	4 months	1,000,000	8,000,000	Salary for 2 research assistants for 4 months	Hiring research assistants	July-August
	1 Data Analyst	4 months	1,000,000	4,000,000	Salary for 1 data analyst for 4 months	Hiring data analyst	July-August
	Field facilitation for 3 research team members.	4 months	200,000	2,400,000	Per diem and accomodation costs for 3 research team members.	Data collection and analysis.	July-October
Data Collection	100 Participants	1 months	50,000	5,000,000	Sitting allowance and travel expenses for participants.	Conducting Surveys and Interviews	August
Design and Development	1 System Developer	1 month	3,000,000	3,000,000	Development and design of the online family property record system	Designing the Online System	September

Materials and Supplies	3 laptops	1 months	5,000,000	15,000,000	Purchasing laptops to store and analyse data as well as design the online system.	Purchasing a laptop.	July - August
	Administrative Costs	4 months	500,000	2,000,000	Costs for printing, field supplies like notebooks and pens, and other materials	Purchasing Materials and Supplies	July - October
	1 Camera	1 month	5,000,000	5,000,000	Purchasing a camera to take pictures for the final project report.	Purchasing a camera.	July-August
	Software purchasing and installation	1 month	5,000,000	5,000,000	Purchasing software packages for design of the online family property record system.	Purchasing and installation of software packages.	August-September
	Maintenance and Repair Costs	4 months	200,000	800,000	Maintenance costs for the laptops and software purchased.	Maintaining and repairing research items.	July-October
Report Writing and Editing	1 Final Research Project Report	1 month	500,000	500,000	Drafting, editing, and printing the final research report	Writing and Submitting Report	October
Miscellaneous	-	4 months	200,000	800,000	Buffer for unforeseen expenses	Contingency Fund	July - October

TOTAL				59,500,000			
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Table 3: Research study budget

Appendix VII: TIME SCOPE

This study is proposed to take a period of 4 months and that is from July 2024 to October 2024.

Month	Activity
July 2024	Submit research proposal.
August 2024	Data Collection and Analysis.
September 2024	Design the online system.
October 2024	Write and submit the research project report.

Appendix VIII: SAMPLE CODE

This is the php and html code from the system file of index.php
(C:\Users\LUNA\Desktop\fps_db files\index.php)

The Family Property Record System database was abbreviated to fps_db for easy management.

```
<?php
session_start();

?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="assets/styles/style.css?v2" >
  <title>Login</title>
</head>
<body>
  <div class="container">
    <h1 class="heading-1">Family Property Keeper</h1>
    <div class="box form-box">
```

```

<?php
    if(isset($_POST['submit']))
    {

        include("php/config.php");

        $email = mysqli_real_escape_string($con, $_POST['email']);
        $password = mysqli_real_escape_string($con,
$_POST['password']);

        //retrieving the hashed password for the given email
        $sql = "SELECT * FROM user WHERE email = '$email'";
        $sql_result = mysqli_query($con, $sql);

        if($sql_result)
        {
            if(mysqli_num_rows($sql_result) > 0)
            {
                $_row = mysqli_fetch_assoc($sql_result);
                $hashed_password = $_row['password'];
                if(password_verify($password, $hashed_password))
                {
                    //login
                    $result = mysqli_query($con,"SELECT * FROM
user WHERE email='$email' AND Password = '$hashed_password'") or die("Select
Error!");

                    $row = mysqli_fetch_assoc($result);
                    if(is_array($row) && !empty($row))
                    {
                        $_SESSION['valid'] = $row['email'];
                        $_SESSION['username'] =
$row['username'];

                        $_SESSION['id'] = $row['id'];
                    }
                    else
                    {
                        echo "<div class='message'>
<p>Wrong Username or
Password</p>

                        </div> <br>";
                        echo "<a href='index.php'><button
class='btn'>Go Back</button></a>";
                    }
                }
            }
            if(isset($_SESSION['valid']))
            {
                header("Location: dashboard.php");
            }
        }
    }
}

```

```

        }
        else
        {
            echo "<div class='message'>
                <p>Incorrect Password</p>
            </div> <br>";
            echo "<a href='index.php'><button
class='btn'>Try Again</button></a>";
        }
    }
    else
    {
        echo "<div class='message'>
            <p>Username Not Found</p>
        </div> <br>";
        echo "<div class='message-box'>
            <a href='javascript:self.history.back()'><button
class='btn'>Try Again</button></a>
            <a href='register.php'><button
class='btn'>Register Now</button></a>
        </div>
        ";
    }

}
else
{
    echo "Error executing query";
}

}
else {

?>

<header>Login</header>
<form action="" method="post">
    <div class="field input">
        <label for="email">Email</label>
        <input type="email" name="email" id="email"
autocomplete="off" required>
    </div>
    <div class="field input">
        <label for="password ">Password</label>
        <input type="password" name="password" id="password"
required>

```

```

        </div>
        <div class="field">
            <input type="submit" class="btn" name="submit"
value="Login" required>
        </div>
        <div class="links">
            Don't have an account? <a href="register.php">Register
Now</a>
        </div>
    </form>
</div>
<?php
    }
?>
</div>

</body>
</html>

```

Figure 3: Sample code for the system, *index.php*